Experiment:-17

Problem Statement:-

In Linux, the wc command is commonly used to count lines, words, and characters in a file. However, understanding how to perform these operations manually using shell scripting helps students gain deeper insight into text processing, looping constructs, and command substitution. This knowledge is particularly useful in automation, log analysis, and cyber security scripting tasks.

The problem is to write a shell script that counts the number of lines, words, and characters in a given input file without using the wc command, using Bash scripting techniques and basic text-processing logic.

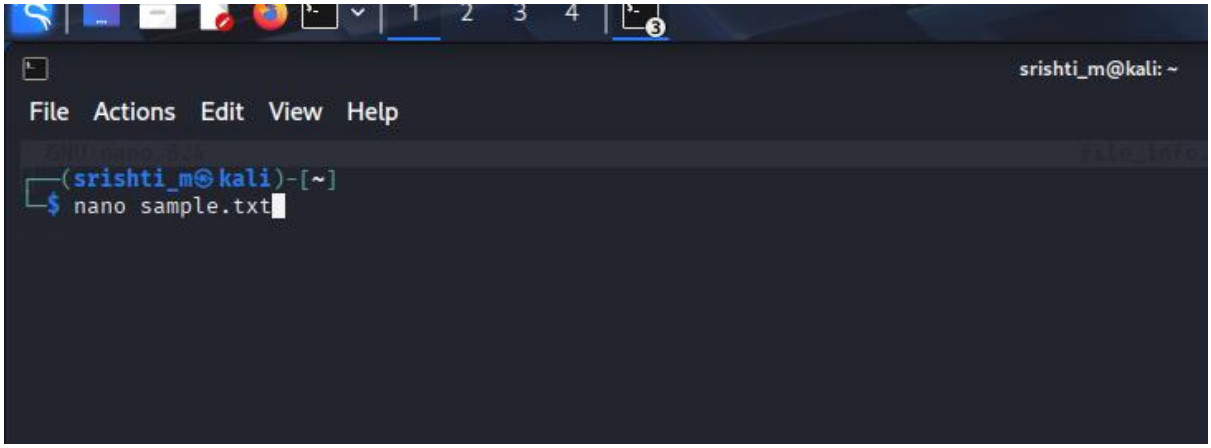
Aim:-Shell script to count line, word and character without using wc

Procedure:-

A.Creating Sample Input File

1. Open the terminal.

2. Create a sample text file: nano sample.txt



3. Enter the following content: Linux is powerful ,Bash scripting is useful ,Cyber Security relies on Linux

```
srishti_m@kali: ~
File  Actions  Edit  View  Help
  GNU nano 8.4                                                    sample.txt
linux is powerful
bash scripting is useful
cyber security relies on linux
```

4. Save and exit the editor.

B. Writing the Shell Script

5. Create a script file: nano count1_text.sh

```
File  Actions  Edit  View  Help
  GNU nano 8.4                                                    sample1_txt.sh *
┌──(srishti_m㉿kali)-[~]
└─$ nano sample.txt

┌──(srishti_m㉿kali)-[~]
└─$ nano count1_text.sh

┌──(srishti_m㉿kali)-[~]
└─$
```

6. Enter the following script:

```
File  Actions  Edit  View  Help
  GNU nano 8.4                                          count1_text.sh
#!/bin/bash
lines=0
words=0
chars=0
while read line
do
    lines=$((lines + 1))
chars=$((chars + ${#line}))
    for word in $line
    do
    words=$((words + 1))
    done
done < sample.txt
echo "number of lines: $lines"
echo "number of words: $words"
echo "number of characters: $chars"
```

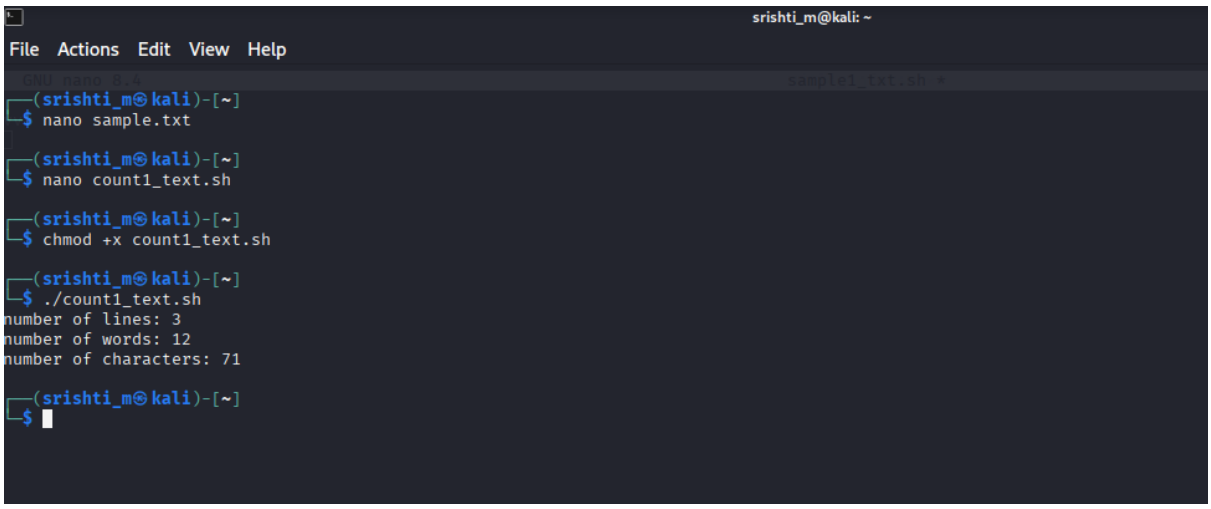
7. Save and exit the editor.

8. Make the script executable: chmod +x count1_text.sh

```
File  Actions  Edit  View  Help
  GNU nano 8.4                                          file_info.sh *
┌──(srishti_m㉿kali)-[~]
└─$ nano sample.txt

┌──(srishti_m㉿kali)-[~]
└─$ nano count_text.sh

┌──(srishti_m㉿kali)-[~]
└─$ chmod +x count_text.sh
```

C. Executing the Script

9. Run the script: ./count1_text.sh